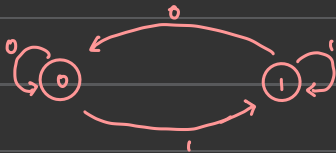


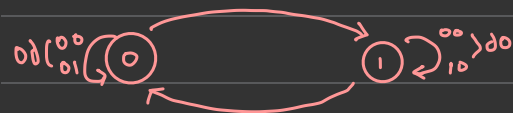
Problem Set 5

1. a. D Flip-Flop



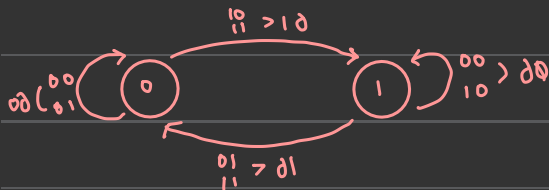
D	Q
0	0
1	1

b. SR flip flop



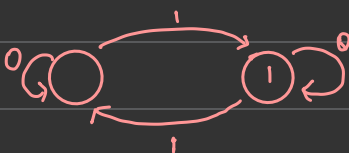
S	R	Q	
0	0	Q	Hold
0	1	0	Reset
1	0	1	Set
1	1	x	

c. JK flip flop



J	K	Q
0	0	Q
0	1	0
1	0	1
1	1	$\bar{Q}$

d. T flip flop



T	Q
0	Q
1	$\bar{Q}$

2. a SR latch

S	R	Q	Q [*]
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	x
1	1	1	x

reset

reset

SR		00	01	11	10
Q	0	0	0	x	1
	1	1	0	x	1

$$\begin{aligned}
 Q_{n+1} &= (SR + SR')(Q_n + Q'_n) + Q_n(SR' + SR) \\
 &= S + QR'
 \end{aligned}$$

b. JK flip flop

J	K	q	q ⁺
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

JK	00	01	11	10
0	0	0	1	1
1	1	0	0	1

$$Q^+ = JQ' + K'Q$$

c. Gated D latch

G	D	q	q ⁺
0	x	0	0
0	x	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

GD	00	01	10	11
0			1	
1	1	1	1	1

$$Q^+ = DG + G'Q$$

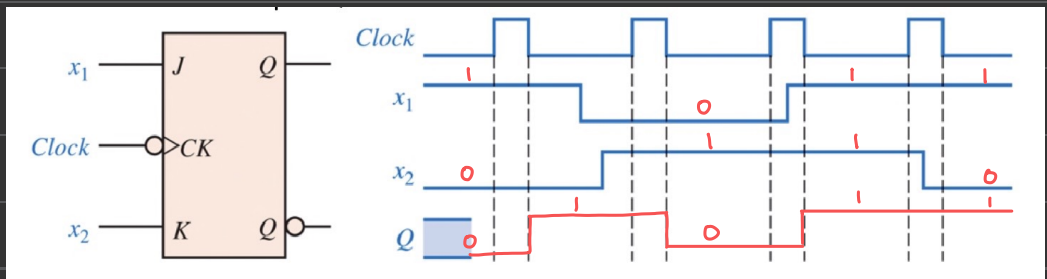
d. T flip flop

T	Q	Q'
0	0	0
0	1	1
1	0	1
1	1	0

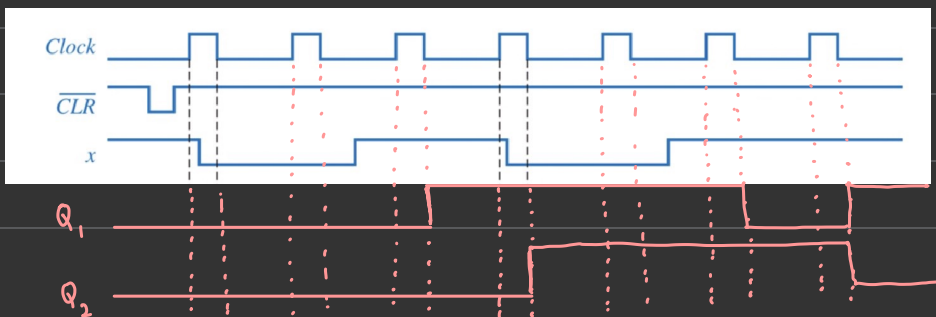
	T	0	1
Q	0	0	1
	1	1	0

$$Q^+ = T'Q + Q'T$$

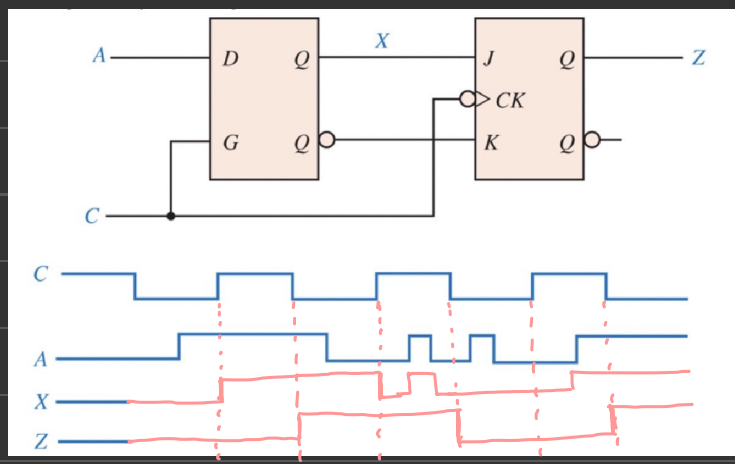
3.



4.



5.



6.

Present state y	Next state Y	JK flip-flop $J \quad K$		D flip-flop D	SR flip-flop $S \quad R$		T flip-flop T
0	0	0	X	0	0	X	0
0	1	1	X	1	1	0	1
1	0	X	1	0	0	1	1
1	1	X	0	1	X	0	0

7.

